

SOFTWARE REQUIREMENTS SPECIFICATION

AutoDezire

AI-Powered Vehicle Recommendation & Ergonomic Decision Platform

Complete UML Modeling & Use Case Specification Suite

Project: AutoDezire Recommendation Platform

Standard: IEEE 830-1998 Software Requirements Specification

Artifacts: Use Case Models • Domain Class Diagrams • System Architecture • Sequence
Flows • Traceability

Date: September 2026 | **Version:** 1.0.0

1. Executive Summary & Problem Domain

AutoDezire is a full-stack, AI-powered vehicle advisory platform engineered to address the critical gap in automobile purchasing: matching a buyer's physical ergonomics, daily commute topography, family passenger needs, and long-term operating budget with the exact vehicular engineering profile. Unlike legacy search engines that rank cars solely on horsepower or generic price points, AutoDezire calculates a weighted 12-dimension deterministic suitability score (safety ratings, ground clearance for bad roads, cabin headroom, seat height/inseam for two-wheelers, fuel economy, and running cost) paired with an interactive OpenRouter Google Gemma 4 26B conversational advisor.

Key Architectural Philosophy: Deterministic Suitability Engine (mathematical guarantee against hallucinations) paired with a Generative LLM Context Engine (OpenRouter Gemma 26B) for nuanced natural language reasoning.

2. System Actors & Role Hierarchy

Actor Name	Actor Type	Responsibilities & Privilege Scope
Guest	Human (Primary)	Unauthenticated user. Allowed to browse vehicle catalogue, filter cars/bikes/scooters, inspect detailed specs, and interact with the AI Advisor demo.
Registered User	Human (Primary)	Authenticated via JWT. Holds persistent 30+ parameter lifestyle profile, bookmarks saved wishlists, receives individualized suitability scores, compares vehicles.
Administrator	Human (Secondary)	Privileged administrative user (verified via admin JWT role). Exercises CRUD control over car and bike catalogs, monitors platform metrics and listing status.
AI Service (Gemma)	Automated External Actor	External OpenRouter LLM (Google Gemma 4 26B). Ingests user lifestyle constraints, vehicle engineering parameters, and outputs personalized advice.

3. High-Level System Overview Use Case Diagram

The System Overview diagram specifies the system boundary of AutoDezire, displaying the four actors and their associations with the primary functional subsystems: Authentication, Vehicle Discovery, AI Advisor, Saved Wishlists, and the Administrative Dashboard.

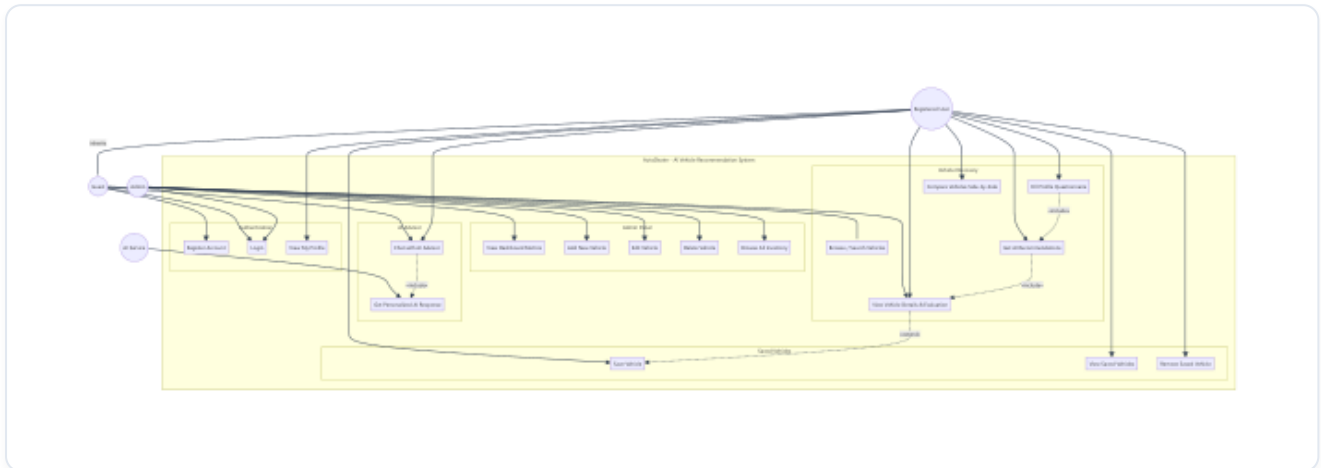


Figure 1: AutoDezire High-Level System Overview Use Case Diagram

4. Detailed Module-Level Use Case Diagrams & Specifications

4.1 Authentication & Identity Module

Manages user registration, credential authentication, JWT token generation, bcrypt password hashing, and seamless fallback to memory storage.

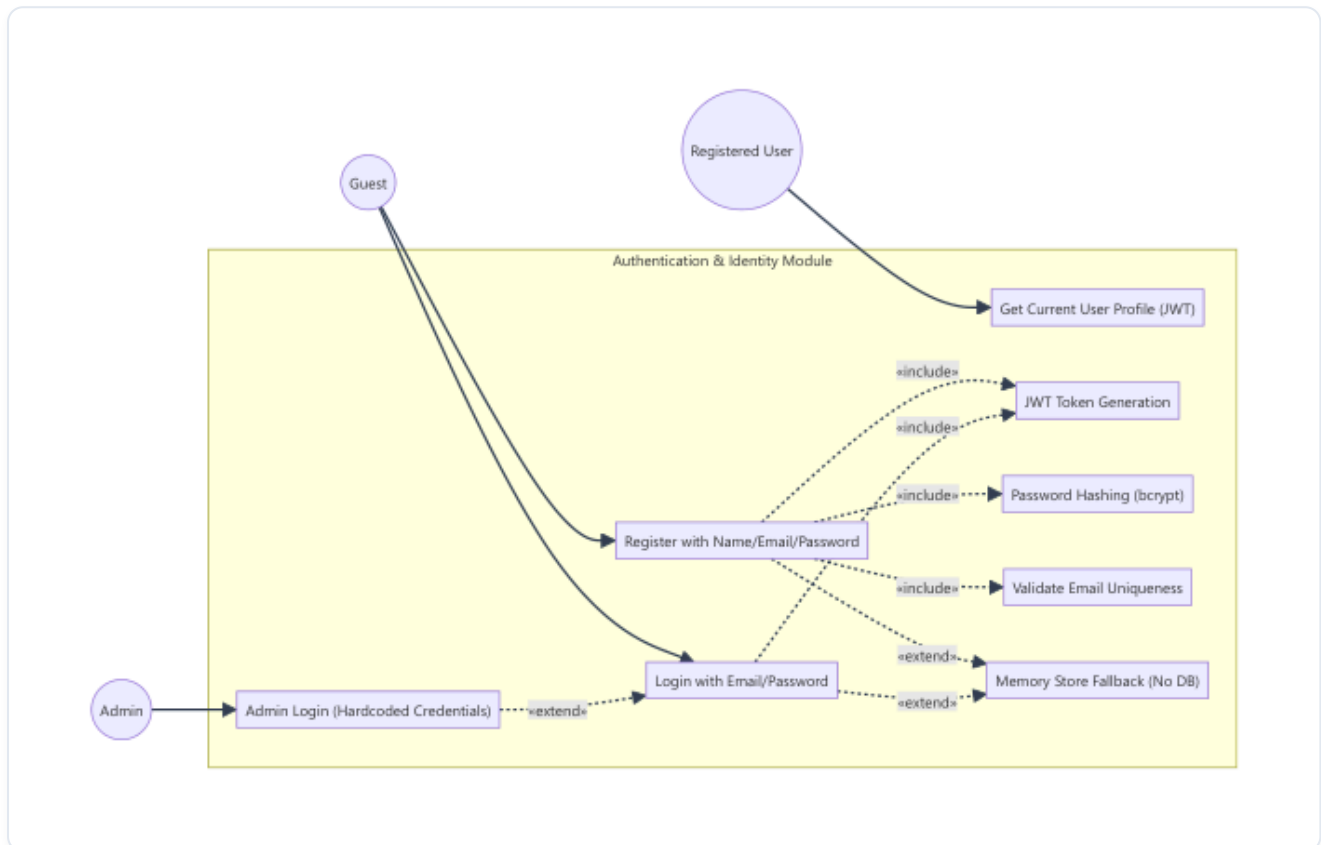


Figure 2: Authentication & Identity Module Use Case

Use Case	Actor	Priority	«include»	«extend»	Functional Summary
UC-01: Register	Guest	High	JWT Token Generation, Password Hashing, Validate Email	Memory Store Fallback	Creates user record with hashed password and returns bearer JWT token.
UC-02: Login	Guest / Admin	High	JWT Token Generation	Admin Hardcoded Override	Authenticates credentials and issues authenticated session token.
UC-03: View Profile	User	Medium	JWT Protection Middleware	—	Retrieves authenticated user profile and lifestyle parameters.

4.2 Vehicle Discovery & Recommendation Engine Module

Gathers 30+ lifestyle parameters through an interactive questionnaire, processes constraints through the 12-dimension suitability engine, and presents ranked, categorized recommendations.

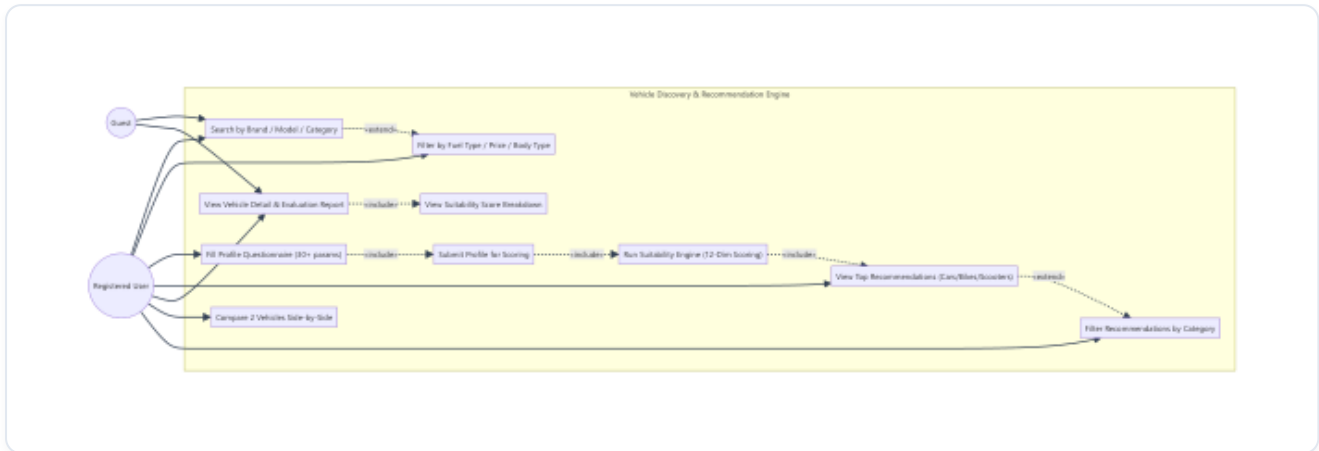


Figure 3: Vehicle Discovery & Recommendation Engine Use Case

Use Case	Actor	Priority	«include»	«extend»	Functional Summary
UC-05: Questionnaire	User	High	Submit Profile for Scoring	—	Captures height, age, daily km, terrain, passengers, luggage, and budget.
UC-06: Recommendations	User	High	12-Dimension Suitability Engine	Filter by Category	Returns ranked list of best-fit vehicles grouped by Cars, Bikes, and Scooters.
UC-07: Vehicle Details	Guest / User	High	Suitability Breakdown	Save to Wishlist	Displays comprehensive specifications, NCAP safety, and score breakdowns.
UC-08: Compare Vehicles	User	Medium	—	—	Renders side-by-side metric comparison between selected models.

4.3 AI Conversational Advisor Module

Leverages OpenRouter Google Gemma 4 26B LLM with dynamic context injection: user physical profile, current vehicle specifications, and suitability breakdown.

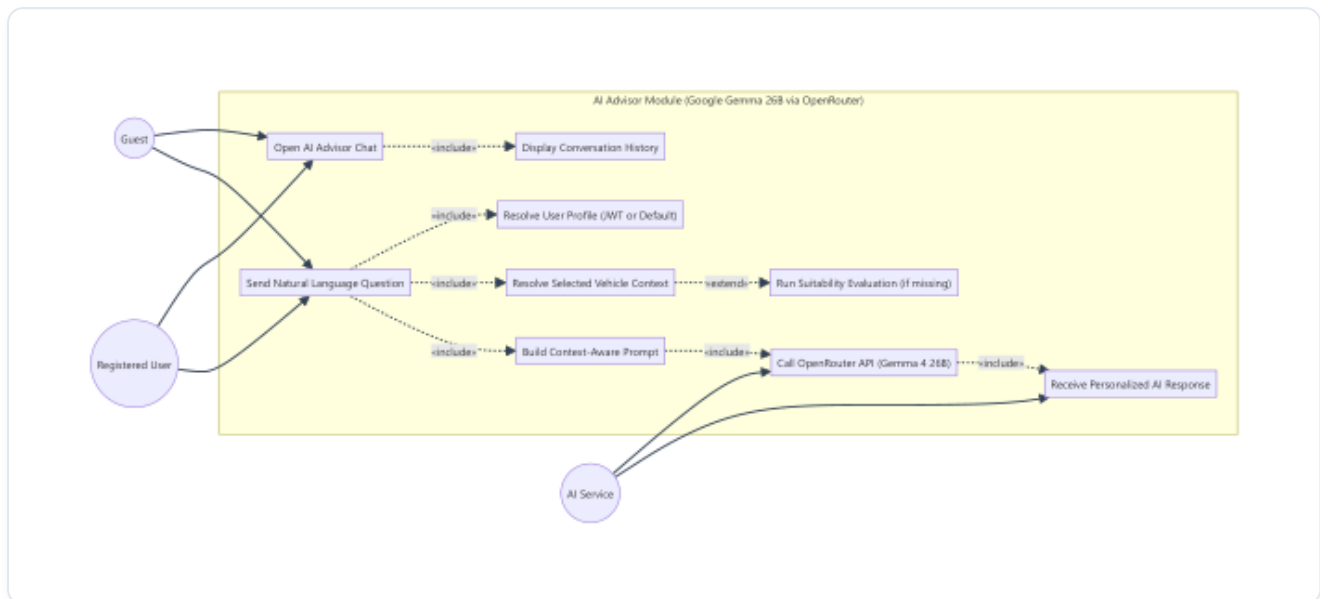


Figure 4: AI Advisor Module Use Case

4.4 Saved Vehicles & Admin Management Modules

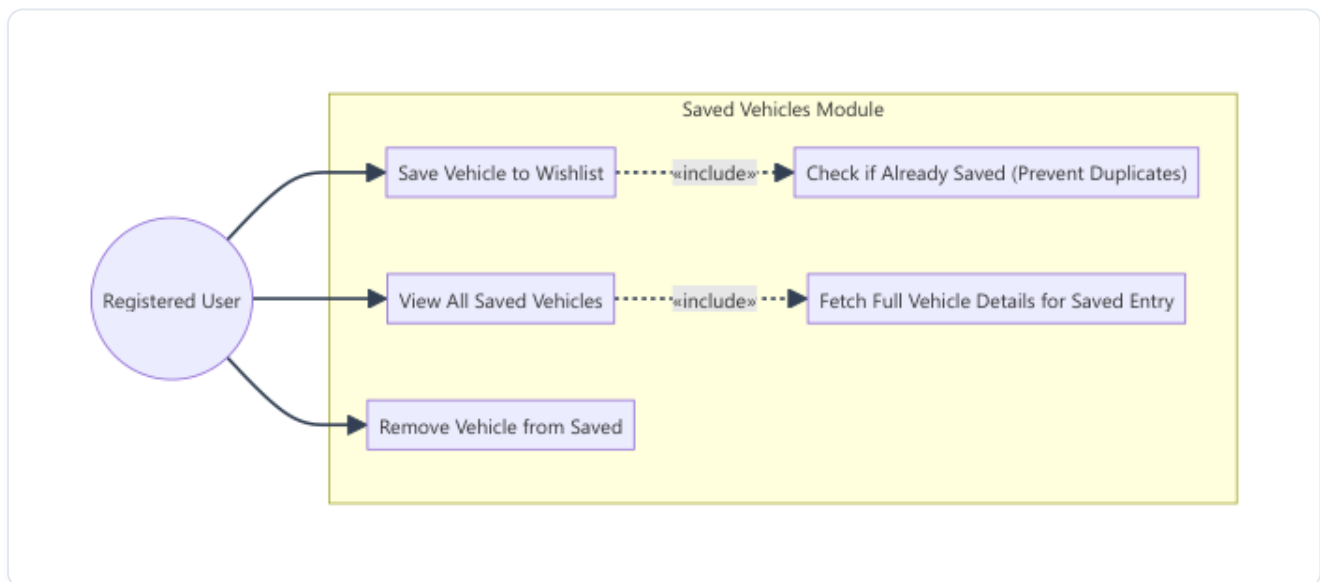


Figure 5: Saved Vehicles Module Use Case

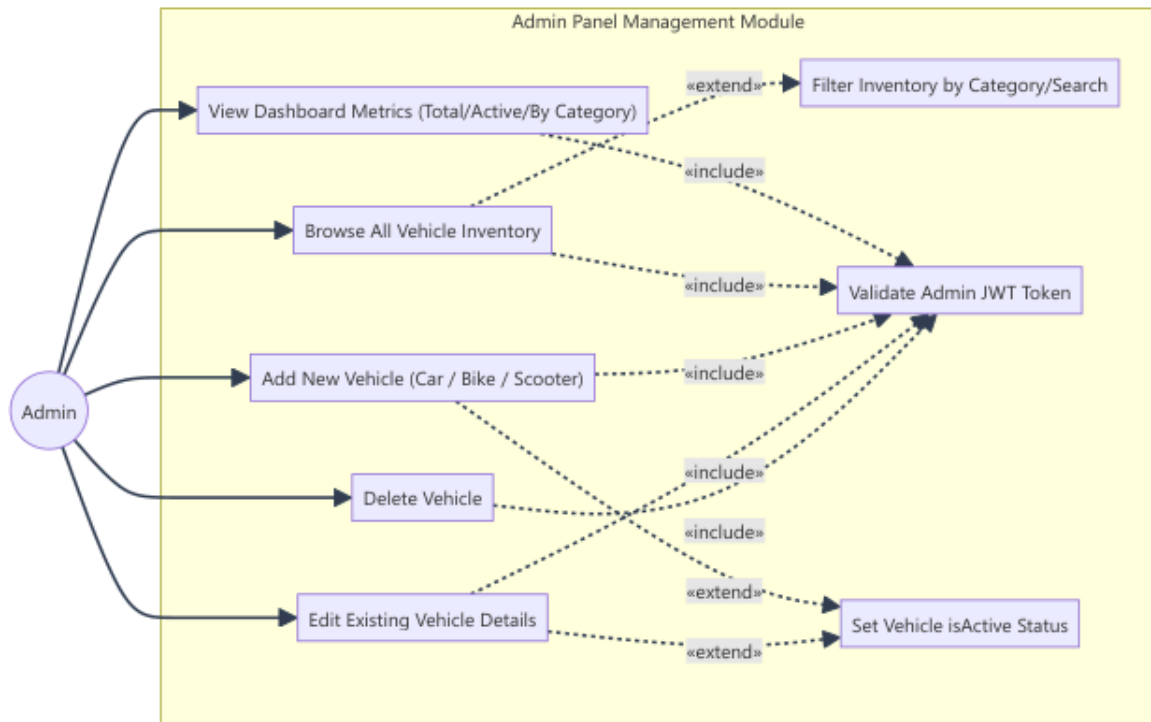


Figure 6: Admin Panel Management Use Case

5. UML Domain Structural Class Diagram

The following class diagram documents the Mongoose ODM domain model hierarchy, embedded documents, and computational objects.

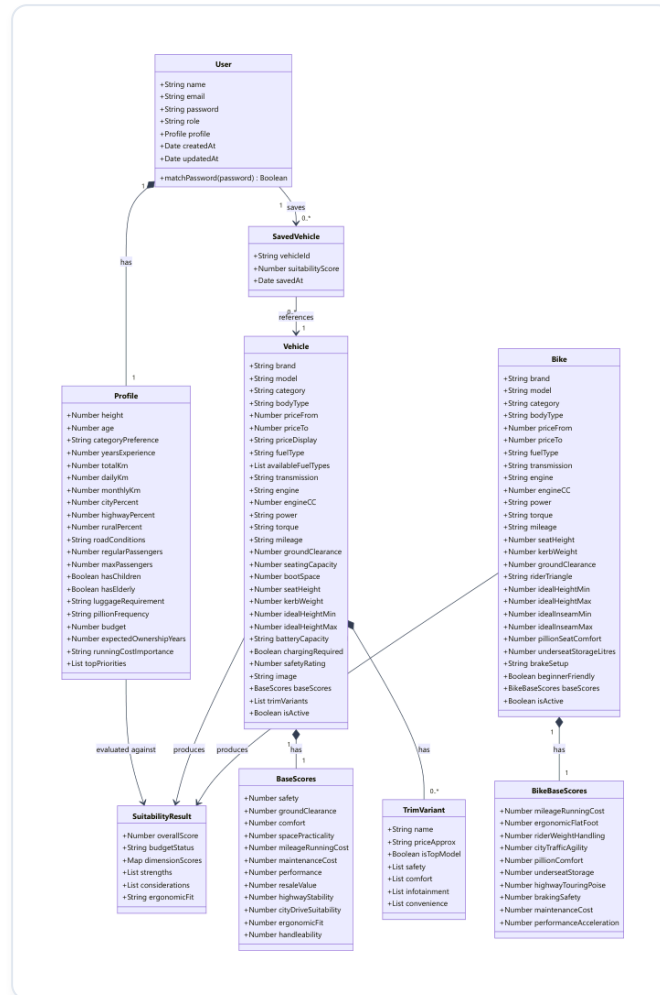


Figure 7: UML Domain Class Diagram

Entity Class	Type	Attributes & Types	Relationships & Associations
User	Mongoose Model	name: String, email: String, password: String, role: Enum, profile: Profile	Composition with Profile (1:1). References SavedVehicle (1:N).
Vehicle	Mongoose Model	brand, model, category, bodyType, priceFrom, priceTo, mileage, groundClearance, safetyRating, baseScores: BaseScores, trimVariants: TrimVariant[]	Composition with BaseScores (1:1). Composition with TrimVariant (1:N).
Bike	Mongoose Model	brand, model, category, seatHeight, kerbWeight, riderTriangle, idealInseamMin, pillionSeatComfort, baseScores: BikeBaseScores	Composition with BikeBaseScores (1:1).
SavedVehicle	Mongoose Model	vehicleId: ObjectId (Ref), suitabilityScore: Number, savedAt: Date	Association with Vehicle.
SuitabilityResult	Compute Object	overallScore: Number, budgetStatus: String, dimensionScores: Map, strengths: String[], considerations: String[]	Computed output of SuitabilityEngine for any Vehicle + Profile.

6. System Architecture & Component Interaction

Visualizes the 4-tier deployment topology: Client Tier (Vite React), Server Tier (Node.js Express), Persistence Tier (MongoDB Atlas), and External AI Tier (OpenRouter Gemma).

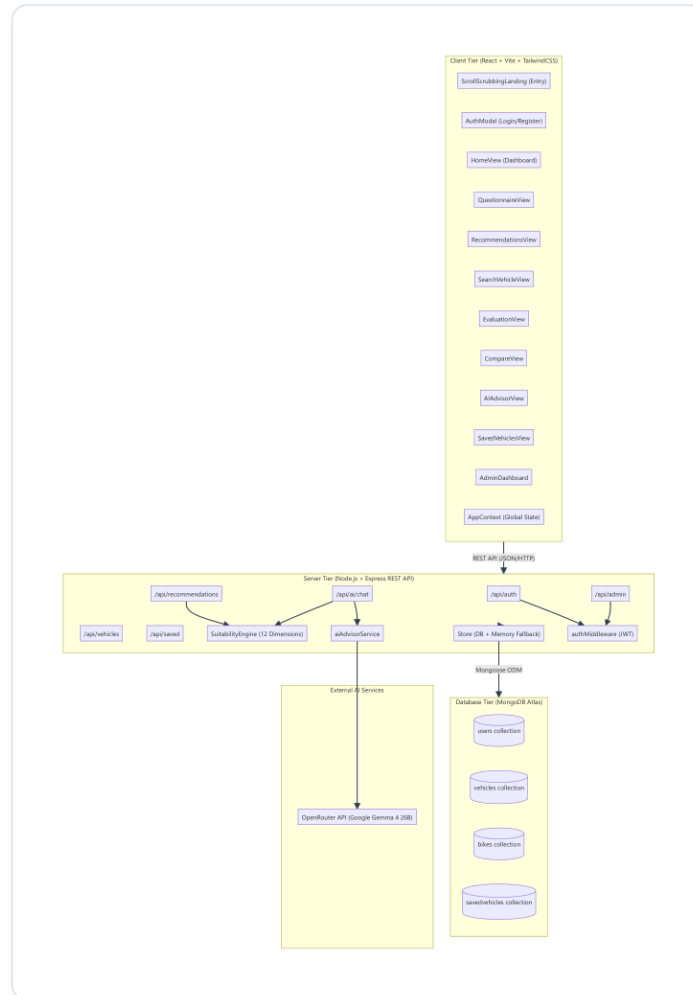


Figure 8: 4-Tier System Architecture & Deployment Diagram

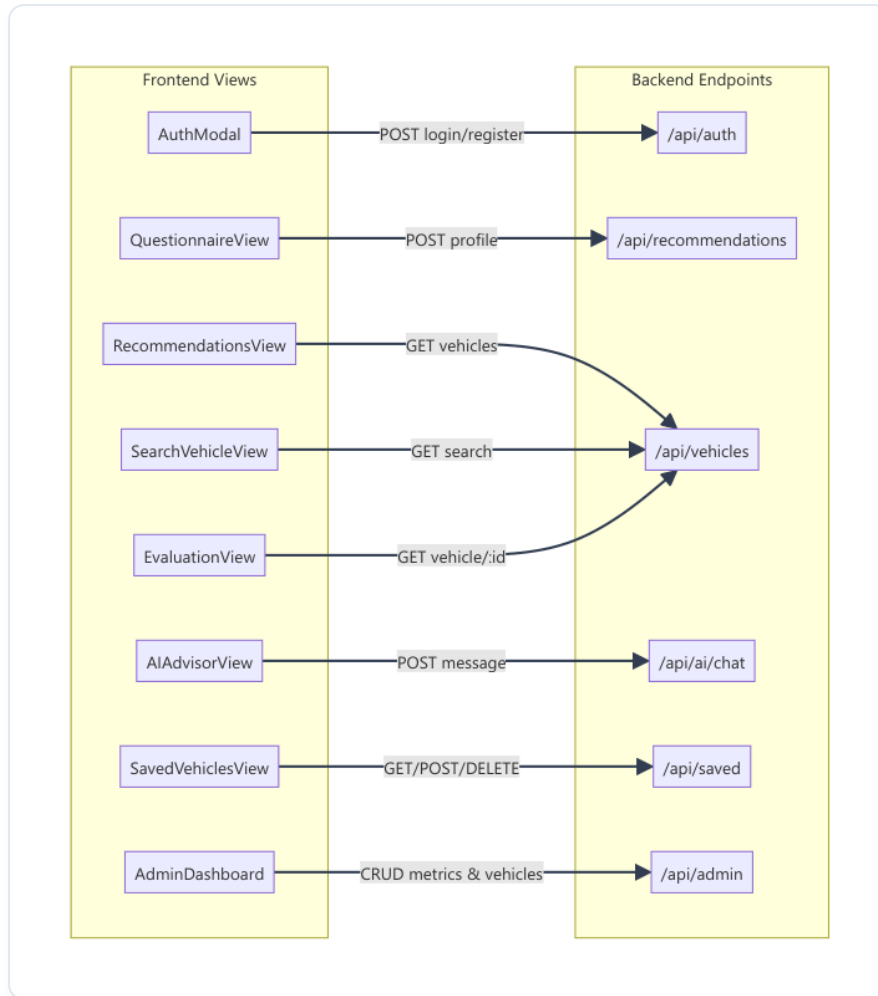


Figure 9: Frontend Components to Express Endpoints Interaction Map

7. Behavioral UML Sequence Diagrams

7.1 AI Advisor Chat Interaction Flow

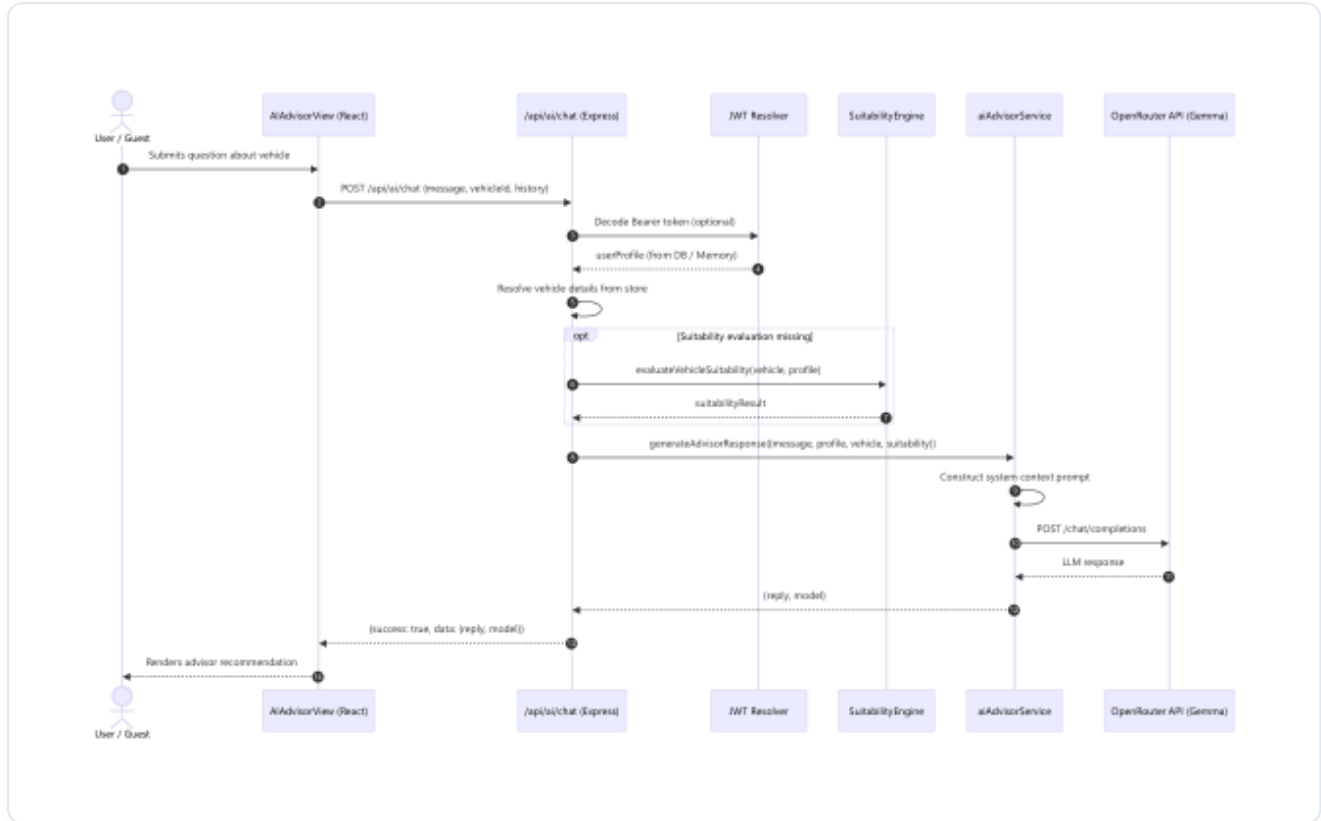


Figure 10: UML Sequence Diagram — AI Advisor Chat Interaction Flow

7.2 Vehicle Recommendation & Scoring Execution Flow

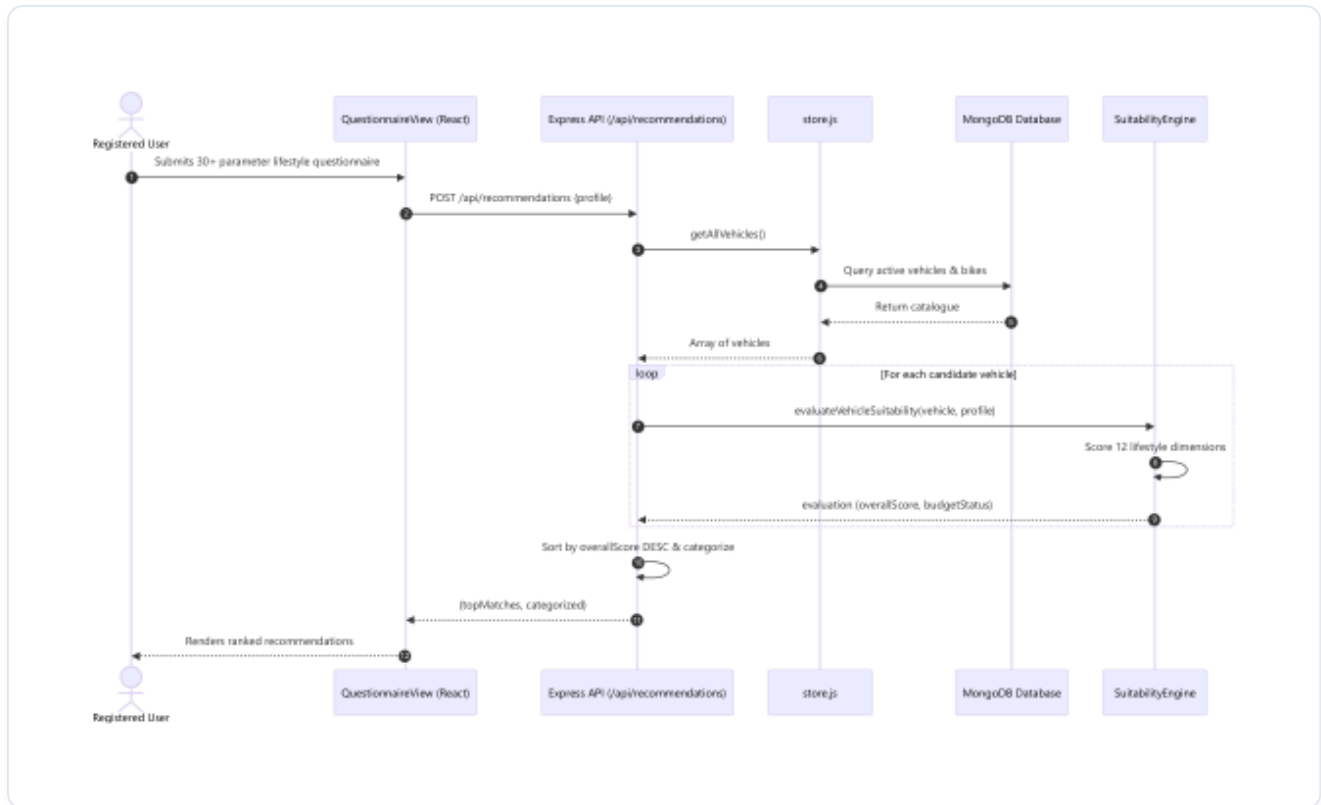


Figure 11: UML Sequence Diagram — Vehicle Recommendation & Scoring Flow

8. Requirements Traceability Matrix (RTM)

Maps functional use cases to backend REST API endpoints and React frontend components.

UC ID	Use Case Title	Actor	REST API Endpoint	Frontend Component
UC-01	Register Account	Guest	POST /api/auth/register	AuthModal.jsx
UC-02	Login	Guest / Admin	POST /api/auth/login	AuthModal.jsx
UC-03	View Profile	User	GET /api/auth/me	SettingsView.jsx
UC-04	Search / Filter Vehicles	Guest / User	GET /api/vehicles	SearchVehicleView.jsx
UC-05	Submit Questionnaire	User	POST /api/recommendations	QuestionnaireView.jsx
UC-06	Get AI Recommendations	User	POST /api/recommendations	RecommendationsView.jsx
UC-07	View Vehicle Details	Guest / User	GET /api/vehicles/:id	EvaluationView.jsx
UC-08	Compare Vehicles	User	Local State / Store	CompareView.jsx
UC-09	Chat with AI Advisor	Guest / User	POST /api/ai/chat	AIAdvisorView.jsx
UC-10	Receive AI Advice	AI Service	OpenRouter API	AIAdvisorView.jsx
UC-11	Save Vehicle Wishlist	User	POST /api/saved	EvaluationView.jsx
UC-12	View Saved Vehicles	User	GET /api/saved	SavedVehiclesView.jsx
UC-13	Remove Saved Vehicle	User	DELETE /api/saved/:id	SavedVehiclesView.jsx
UC-14	View Admin Metrics	Admin	GET /api/admin/metrics	AdminDashboard.jsx
UC-15	Browse Admin Inventory	Admin	GET /api/admin/vehicles	AdminDashboard.jsx
UC-16	Add Vehicle	Admin	POST /api/admin/vehicles	AdminDashboard.jsx
UC-17	Edit Vehicle	Admin	PUT /api/admin/vehicles/:id	AdminDashboard.jsx
UC-18	Delete Vehicle	Admin	DELETE /api/admin/vehicles/:id	AdminDashboard.jsx
UC-19	12-Dim Suitability Engine	System Engine	suitabilityEngine.js	Backend Internal
UC-20	Interactive Video Scrub	Guest	Client State	ScrollScrubbingLanding.jsx